WORKSHEET

Areas using rectangles

1 Find an approximation for $\int_0^4 2x + 1\, dx$ using:

a left rectangles of width 1 unit

b right rectangles of width 1 unit

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c centred rectangles of width 1 unit.

2 Find an approximation for $\int_0^2 x^2\,dx$ using left, right and centred rectangles of width:

a 1 unit

b 0.5 units

c 0.4 units.

3 Use left rectangles 0.5 units wide to find approximations for:

a $\int_1^4 -x^2 + 20 \, dx$

b $\int_3^4 x^3 \, dx$

c $\int_0^1 -x^2 - 2x + 6 \, dx$

d $\int_1^3 2^x dx$

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4 Use right rectangles 0.4 units wide to find approximations for:

a $\int_0^2 \sin(x)\ dx$

b $\int_{0.4}^{1.6} \frac{1}{x}\ dx$

c $\int_{14}^{16} \sqrt{x}\ dx$

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d $\int_0^{0.8} e^{2x}\,dx$

5 Use centre rectangles 0.2 units wide to find approximations for:

a $\int_0^{0.8} \cos(2x)\,dx$

b $\int_{1}^{1.6} \frac{2-x}{x+1}$

c $\int_{22.3}^{22.9} \sqrt{x+5}$

d $\int_3^4 x^3 - 2x^2 + 3x + 1$

6 **CAS** Use a CAS calculator or spreadsheet to find approximations for the following.

a $\int_0^5 x^2 dx$ using 20 left rectangles

b $\int_0^{\pi} 2 + \sin(3x)\ dx$ using right rectangles of width $\frac{\pi}{40}$ units.

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Answers

1 **a** 16 units^2

b 24 units^2

c 20 units^2

2 **a** Left $\approx$ 1 unit^2; right $\approx$ 5 units^2; centre $\approx$ 2.5 units^2

b Left $\approx$ 1.75 units^2; right $\approx$ 3.75 units^2; centre $\approx$ 2.625 units^2

c Left $\approx$ 1.92 units^2; right $\approx$ 3.52 units^2; centre $\approx$ 2.64 units^2

3 **a** $\approx$ 42.625 units^2

b $\approx$ 34.9375 units^2

c $\approx$ 5.375 units^2

d $\approx$ 7.2426 units^2

4 **a** $\approx$ 1.5791 units^2

b $\approx$ 1.0833 units^2

c $\approx$ 7.7961 units^2

d $\approx$ 2.8714 units^2

5 **a** $\approx$ 0.5031 units^2

b $\approx$ 0.1866 units^2

c $\approx$ 3.1521 units^2

d 29.835 units^2

6 **a** $\approx$ 38.5938 units^2

b $\approx$ 6.9468 units^2

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